Assignment 7

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Please note that

- **Released date: April 12, Friday.**
- **Due date: April 24, Wednesday, by 11:55 pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "123010XXX ASS6.pdf" (Your student ID + ASS No.). You may either typeset your answers directly using computers or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size to 10MB.)

Question 1. Find the constants a and b such that the vectors $u = [a, 4, -b]^T$ and $v = [a, 1, b]^T$ are orthogonal and $a = b + 1$.

Solution

Vectors u and v are orthogonal, hence their inner product is equal to zero

$a^2 + 4 - b^2 = 0$.

Given that $a = b + 1$, substitute a by $b + 1$ in the above equation

$(b + 1)^2 + 4 - b^2 = 0$

Expand the above equation and simplify

$2b + 5 = 0$.

Solve for b

$b = -\frac{5}{2}$

Find a

$a = b + 1 = -\frac{5}{2} + 1 = -\frac{3}{2}$

Hence, the vectors u and v are orthogonal if $a = -\frac{3}{2}$ and $b = -\frac{5}{2}$.

Question 2. If $x = (x_1, x_2)^T$,$y = (y_1, y_2)^T$,and $z = (z_1, z_2)^T$ are arbitrary vectors in R^2, prove that:

1. $x^T x \geqslant 0$
2. $x^T y = y^T x$
3. $x^T(y + z) = x^T y + x^T z$

Solution

1. $x^T x = x_1^2 + x_2^2 \geqslant 0$.

2. $x^T y = x_1 y_1 + x_2 y_2 = y^T x$.

3. $x^T(y+z) = x_1 y_1 + x_1 z_1 + x_2 y_2 + x_2 z_2 = x^T y + x^T z$.

Question 3. Let $x = (4, 4, -4, 4)^T$ and $y = (4, 2, 2, 1)^T$.

1. Determine the angle between x and y.
2. Determine the distance between x and y.

Solution

1. To find the angle between two vectors x and y, we can use the dot product formula:

   $cos\theta = \frac{x^T y}{||x||||y||}$

   Using the given vectors, we have:

   $x^T y = (4)(4) + (4)(2) + (-4)(2) + (4)(1) = 20$

   $||x|| = \sqrt{4^2 + 4^2 + (-4)^2 + 4^2} = \sqrt{64} = 8$

   $||y|| = \sqrt{4^2 + 2^2 + 2^2 + 1^2} = \sqrt{25} = 5$

   Substituting these values into the formula for $cos\theta$, we get:

   $cos\theta = \frac{20}{(8)(5)} = \frac{1}{2}$

   Taking the inverse cosine of both sides, we get:

   $\theta = cos^{-1}(\frac{1}{2}) = 60^\circ$

   Therefore, the angle between x and y is 60°

2. $||x - y|| = \sqrt{0^2 + 2^2 + (-6)^2 + 3^2} = \sqrt{49} = 7$

   Therefore, the distance between x and y is 7.

Question 4. For the following pair of vectors x and y, find the vector projection p of x onto y and verify that p and $x - p$ are orthogonal: $x = (3, 5)^T$,$y = (1, 1)^T$

Solution

The vector projection of x onto y is
$p = \frac{x^T y}{y^T y} y = (4,4)^T$
Therefore,$x - p = (3,5)^T - (4,4)^T = (-1,1)^T$. It is easy to verify that $p^T(x-p) = 0 \Rightarrow p$ and $x - p$ are orthogonal.

Question 5. State whether each of the following is true or false:

a) If the subspaces V and W are orthogonal, then $V^\perp$ and $W^\perp$ are also orthogonal.

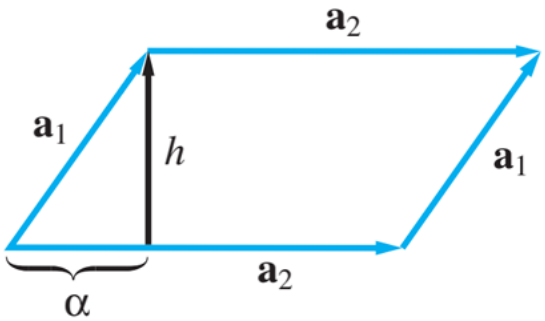
b) If V is orthogonal to W and W orthogonal to Z then V is orthogonal to Z.

Solution

a) In $\mathbb{R}^n$, suppose that $V = W = \{0\}$. Then since the zero vector is orthogonal to itself, V is orthogonal to $W = V$. However we have $V^\perp = W^\perp = \mathbb{R}^n$and $\mathbb{R}^n$ is not orthogonal to itself, so $V^\perp$ and $W^\perp$ are not orthogonal. The statement is false.

b) Suppose that $V = Z = \mathbb{R}^n$ and $W = \{0\}$. Then V is orthogonal to W and W is orthogonal to Z but V is not orthogonal to Z because $\mathbb{R}^n$ is not orthogonal to itself. The statement is false.

Question 6. Let A be a 2×2 matrix with linearly independent column vectors a_1 and a_2. If a_1 and a_2 are used to form a parallelogram P with altitude h (see the figure). show that $h^2||a_2||^2 = ||a_1||^2||a_2||^2 - (a_1^T a_2)^2$.



Solution

By the Pythagorean Theorem
$\alpha^2 + h^2 = ||a_1||^2$
where α is the scalar projection of a_1 into a_2. It follows that
$\alpha^2 = \frac{(a_1^T a_2)^2}{||a_2||^2}$

and
$h^2 = ||a_1||^2 - \frac{(a_1^T a_2)^2}{||a_2||^2}$
Hence
$h^2||a_2||^2 = ||a_1||^2||a_2||^2 - (a_1^T a_2)^2$

Question 7. For each of the following systems $Ax = b$, find all least squares solutions:

1. $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{pmatrix}$, $b = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$
2. $A = \begin{pmatrix} 1 & 1 & 3 \\ -1 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}$,$b = \begin{pmatrix} -2 \\ 0 \\ 8 \end{pmatrix}$

Solution

Definition: If A is an $m \times n$ matrix of rank n, the normal equations
$A^T Ax = A^T b$
have a unique solution
$x = (A^T A)^{-1} A^T b$
and x is the unique least squares solution of the system $Ax = b$

1. Let $x = [x_1, x_2]^T$

   Substitute the values of A, x, and b in $A^T Ax = A^T b$

   $$\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix}^T \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix}^T \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

   When solving this linear system, we can obtain

   $x_1 + 2x_2 = 1$

   Note that there are two variables (namely, x_1, x_2) and one equation. So the solution of it must contain 1 variable as free variable.

   Suppose x_2 is the free variable. That is, $x_2 = \alpha, \alpha \in \mathbb{R}$.

   Then from the equation $x_1 + 2x_2 = 1$ get $x_1 = 1 - 2\alpha$.

   Hence the least squares solution is, $(x_1, x_2)^T = \{(1 - 2\alpha, \alpha)^T | \alpha \in \mathbb{R}\}$

2. Similar to (a), we can solve the linear system to obtain:

   $x_1 + 2x_3 = 2$

$x_2 + x_3 = 1$

Note that there are three variables (namely, x_1, x_2, x_3) and two equations. So, the solution of it must contain 1 variable as a free variable.

Suppose x_3 is the free variable.

That is, $x_3 = \alpha, \alpha \in \mathbb{R}$.

Then from the equation $x_2 + x_3 = 1$ get $x_2 = 1 - \alpha$ from the equation $x_1 + 2x_3 = 2$ get $x_1 = 2 - 2\alpha$.

Hence the least squares solution is, $(x_1, x_2, x_3)^T = \{(2-2\alpha, 1-\alpha, \alpha)^T | \alpha \in \mathbb{R}\}$

Question 8. Let $P = A(A^TA)^{-1}A^T$, where A is an $m \times n$ matrix of rank n.

1. Show that $P^2 = P$.
2. Prove that $P^k = P$ for $k = 1, 2, ...$
3. Show that P is symmetric. [Hint: If B is non-singular, then $(B^{-1})^T = (B^T)^{-1}$.

Solution

1. $P^2 = A(A^TA)^{-1}A^TA(A^TA)^{-1}A^T$

   $P^2 = A[(A^TA)^{-1}A^TA](A^TA)^{-1}A^T$

   $P^2 = A(A^TA)^{-1}A^T$

   $P^2 = P$

2. The objective is to prove that $P^k = P$ for $k = 1, 2, ...$ We prove this by using induction on k. If $k = 1$ then we have $P^1 = P$ Assume $P^m = P$, for some m. Consider $P^{m+1} = PP^m = PP = P^2 = P$.

   Hence the result is true by induction on k.

3. The objective is to show that P is symmetric. It is enough to prove $P^T = P$.

   Consider $P^T = [A(A^TA)^{-1}A^T]^T$

$= (A^T)^T[(A^TA)^{-1}]^TA^T$

$= A[(A^TA)^T]^{-1}A^T$

$= A[A^T(A^T)^T]^{-1}A^T$

$= A[(A^TA)]^{-1}A^T$

$= P$

Therefore, P is symmetric.

Question 9. Which of the following sets of vectors form an orthonormal basis for $\mathbb{R}^2$?

(a) $\{(1,0)^T, (0,1)^T\}$

(b) $\{(\frac{3}{5}, \frac{4}{5})^T, (\frac{5}{13}, \frac{12}{13})^T\}$

(c) $\{(1,-1)^T, (1,1)^T\}$

(d) $\{(\frac{\sqrt{3}}{2}, \frac{1}{2})^T, (-\frac{1}{2}, \frac{\sqrt{3}}{2})^T\}$

Solution

(a) Take $v_1 = (1,0)^T$, $v_2 = (0,1)^T$ The inner product of these vectors is calculated as follows:

$< v_1, v_2 >= v_2^T v_1$

$= (0,1)(1,0)^T$

$= 0$

Therefore, the inner product of the vectors v_1 and v_2 is $< v_1, v_2 >= 0$

Find the inner product with the same vectors.

$< v_1, v_1 >= v_1^T v_1 = 1$

$< v_2, v_2 >= v_2^T v_2 = 1$

Since the inner products $< v_1, v_2 >= 0$, $< v_1, v_1 >= 1$, $< v_2, v_2 >= 1$, so by the definition of an orthonormal set, $\{(1,0)^T, (0,1)^T\}$ is an orthonormal.

And, no vector of the set $\{(1,0)^T, (0,1)^T\}$ is a scalar multiple of the other. so it is a linearly independent set and hence it is an orthonormal basis in $\mathbb{R}^2$.

(b) The inner product of these vectors is calculated as follows:

$< v_1, v_2 >= v_2^T v_1$

$= (\frac{5}{12}, \frac{12}{13})(\frac{3}{5}, \frac{4}{5})^T$
$= \frac{63}{65}$
$\neq 0$
Since the inner product of $v_1 = (\frac{3}{5}, \frac{4}{5})^T$, $v_2 = (\frac{5}{13}, \frac{12}{13})^T$ is not equal 0, that is $< v_1, v_2 > \neq 0$. Therefore, by the definition of an orthonormal set, the set $\{(\frac{3}{5}, \frac{4}{5})^T, (\frac{5}{13}, \frac{12}{13})^T\}$ is not an orthonormal. Hence, the set $\{(\frac{3}{5}, \frac{4}{5})^T, (\frac{5}{13}, \frac{12}{13})^T\}$ is not an orthonormal basis for $\mathbb{R}^2$.

(c) The inner product is 0, and the inner product with the same vectors is calculated as follows:
$< v_1, v_1 >= v_1^T v_1 = 2 \neq 1$
Since the inner product $< v_1, v_1 > \neq 1$, so by the definition of an orthonormal set, the set $\{(1,0)^T, (0,1)^T\}$ is not an orthonormal.
Hence, the set $\{(1,-1)^T, (1,1)^T\}$ is not an orthonormal basis for $\mathbb{R}^2$.

(d) Similarly, we can get that this vector set is an orthonormal basis in $\mathbb{R}^2$.

Question 10. $u_1 = (\frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, -\frac{4}{3\sqrt{2}})^T$, $u_2 = (\frac{2}{3}, \frac{2}{3}, \frac{1}{3})^T$, $u_3 = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0)^T$.

(a) Show that $\{u_1, u_2, u_3\}$ is an orthonormal basis for $\mathbb{R}^3$.

(b) Let $x = (1,1,1)^T$. Write x as a linear combination of u_1, u_2 and u_3, and compute $||x||$.

Solution

(a) Consider
$u_1^T u_1 = 1$, $u_2^T u_2 = 1$, $u_3^T u_3 = 1$, $u_1^T u_2 = 0$, $u_1^T u_3 = 0$, $u_2^T u_3 = 0$.
Therefore, the set $\{u_1, u_2, u_3\}$ is an orthonormal basis for $\mathbb{R}^3$.

(b) Let $x = (1,1,1)^T x$ as the linear combination of u_1, u_2 and u_3 is given by $x = c_1 u_1 + c_2 u_2 + c_3 u_3$, where
$c_1 = x^T u_1 = \frac{-\sqrt{2}}{3}$
$c_2 = x^T u_2 = \frac{5}{3}$
$c_3 = x^T u_3 = 0$
Therefore $x = \frac{-\sqrt{2}}{3} u_1 + \frac{5}{3} u_2$, $||x|| = \sqrt{3}$

Question 11. Let u_1 and u_2 form an orthonormal basis for $\mathbb{R}^2$ and let u be a unit vector in $\mathbb{R}^2$. If $u^T u_1 = \frac{1}{2}$, determine the value of $|u^T u_2|$.

Solution

Let u_1 and u_2 form an orthonormal basis for $\mathbb{R}^2$ and let u be a unit vector in $\mathbb{R}^2$.

If $c_1 = u^T u_1 = \frac{1}{2}$ and $c_2 = u^T u_2$, then by the theorem let $\{u_1, u_2, ..., u_n\}$ be an orthonormal basis for an inner product space v. If $v = \sum_{i=1}^{n} c_i u_i$ then $c_i =< v, u_i >$

We have $u = c_1 u_1 + c_2 u_2$. It follows from Parseval's formula that

$1 = ||u||^2$

$= c_1^2 + c_2^2$

$= \frac{1}{4} + c_2^2$

$\Rightarrow \frac{3}{4} = c_2^2$

Hence $|u^T u_2| = |c_2| = \frac{\sqrt{3}}{2}$.

Question 12. Let $\{u_1, u_2, u_3\}$ be an orthonormal basis for an inner product space V. If $x = c_1 u_1 + c_2 u_2 + c_3 u_3$ is a vector with the properties $||x|| = 5$, $< u_1, x >= 4$ and $x \perp u_2$, then what are the possible values of c_1, c_2, c_3?

Solution

$||x||^2 =< c_1 u_1 + c_2 u_2 + c_3 u_3, c_1 u_1 + c_2 u_2 + c_3 u_3 >$

$= \sum_{i=1}^{3} \sum_{j=1}^{3} < c_i u_i, c_j u_j >$

$= \sum_{i=1}^{3} < c_i u_i, c_i u_i >$

$= \sum_{i=1}^{3} c_i^2$

$= c_1^2 + c_2^2 + c_3^2$

If $\{u_1, u_2, u_3\}$ be an orthonormal basis for an inner product space V.

Consider the theorem:

$x = \sum_{i=1}^{3} c_i u_i, c_i =< x, u_i >, i = 1, 2, 3.$

$\therefore c_1 =< x, u_1 >=< u_1, x >= 4$

$c_2 =< x, u_2 >= 0$ as $x \perp u_2$

$c_1^2 + c_2^2 + c_3^2 = 4^2 + 0^2 + c_3^2 = ||x||^2 = 25, c_2^2 = 9$

$c_3 = \pm 3.$

Question 13. For each of the following, use the Gram-Schmidt process to find an orthonormal basis for Row(A)

(a) $A = \begin{pmatrix} -1 & 3 \\ 1 & 5 \end{pmatrix}$

(b) $A = \begin{pmatrix} 2 & 5 \\ 1 & 10 \end{pmatrix}$

Solution

(a) $\{(-\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}})^T, (\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}})^T\}$

(b) $\{(\frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}})^T, (-\frac{5}{\sqrt{29}}, \frac{2}{\sqrt{29}})^T\}$.

Question 14. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}$ and $b = \begin{pmatrix} 12 \\ 6 \\ 18 \end{pmatrix}$

(a) Use the Gram-Schmidt process to find an orthonormal basis for the column space of A.

(b) Factor A into a product QR, where Q has an orthonormal set of column vectors and R is upper triangular.

(c) Solve the least squares problem $Ax = b$.

Solution

(a) The orthonormal basis is

$\{(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})^T, (\frac{-\sqrt{2}}{6}, \frac{4\sqrt{2}}{6}, \frac{-\sqrt{2}}{6})^T\}$

$= \{\frac{1}{3}(2, 1, 2)^T, \frac{\sqrt{2}}{6}(-1, 4, -1)^T\}$

(b) We can factorize A into the product QR, where

$R = \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix} = \begin{bmatrix} 3 & \frac{5}{3} \\ 0 & \frac{\sqrt{2}}{3} \end{bmatrix}$

$Q = \begin{bmatrix} \frac{2}{3} & \frac{-\sqrt{2}}{6} \\ \frac{1}{3} & \frac{2\sqrt{2}}{3} \\ \frac{2}{3} & \frac{-\sqrt{2}}{6} \end{bmatrix}$

(c) $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 12 \\ 6 \\ 18 \end{bmatrix}$

The normal equations for this system are $A^TAx = A^Tb$

$\Rightarrow A^TAx = \begin{bmatrix} 9 & 5 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 66 \\ 36 \end{bmatrix} = A^Tb$

$9x_1 + 5x_2 = 66$

$5x_1 + 3x_2 = 36$

Multiply the first equation with 5 and the second equation with 9 and then subtract.we get $x_1 = 9, x_2 = -3$. Therefore $x = \begin{bmatrix} 9 \\ -3 \end{bmatrix}$.

Question 15. Let V be an inner product space over $\mathbb{R}$, $\forall u, v \in V$, show the following:

(a) $< u, v >= \frac{1}{4}||u + v||^2 - \frac{1}{4}||u - v||^2$

(b) $||u+v||^2 + ||u-v||^2 = 2||u||^2 + 2||v||^2$

Solution

(a) $\frac{1}{4}||u+v||^2 - \frac{1}{4}||u-v||^2$

$= \frac{1}{4}(< u+v, u+v > - < u-v, u-v >)$

$= \frac{1}{4}(< u+v, u > + < u+v, v > -(< u-v, u > - < u-v, v >))$

$= \frac{1}{4}(< u, u > + < v, u > + < u, v > + < v, v > - < u-v, u > + < u-v, v >)$

$= \frac{1}{4}(< u, u > + < v, u > + < u, v > + < v, v > - < u, u > + < v, u > + < u, v > - < v, v >)$

$=< v, u >$

(b) $||u+v||^2 + ||u-v||^2$

$=< u+v, u+v > + < u-v, u-v >$

$=< u, u > + < u, v > + < v, u > + < v, v > + < u, u > - < u, v > - < v, u > + < v, v >$

$= 2 < u, u > +2 < v, v >$

$= 2||u||^2 + 2||v||^2$

Question 16. Let Q be an orthogonal matrix and let $d = det(Q)$. Show that $|d| = 1$.

Solution

$Q^TQ = QQ^T = I$

$det(QQ^T) = det(I) = 1$

$1 = det(Q^TQ) = det(Q^T)det(Q) = det(Q)det(Q) = (det(Q))^2 = 1$

$det(Q) = \pm 1$

$|d| = |det(Q)| = 1$

Question 17. Given the vector space $C[-1,1]$ with inner product

$< f, g >= \int_{-1}^{1} f(x)g(x)dx$

and norm

$||f|| = (< f, f >)^{\frac{1}{2}}$

(a) Show that the vectors 1 and x are orthogonal.

(b) Compute $||1||$ and $||x||$.

Solution

(a) $< 1, x >= \int_{-1}^{1} x dx = 0$

therefore $1 \perp x$.

(b) $||1|| = \sqrt{< 1, 1 >} = (\int_{-1}^{1} 1^2 dx)^{\frac{1}{2}} = \sqrt{2}$

$||x|| = \sqrt{< x, x >} = (\int_{-1}^{1} x^2 dx)^{\frac{1}{2}} = \sqrt{\frac{2}{3}}$

Question 18. (Prove for the Cauchy—Schwarz inequality)

Let V be an inner product space, $\forall u, v \in V$, show that $| < u, v > | \leqslant ||u||||v||(\star)$.

Solution

$\forall t \in \mathbb{R}, ||tv + u||^2 \geqslant 0$

$||tv + u||^2 =< tv + u, tv + u >= t^2||v||^2 + 2t < u, v > +||u||^2 \geqslant 0, \forall t \in \mathbb{R}$

when $v = 0$, $(\star)$ is valid.

when $v \neq 0$, $||v|| \neq 0$, $t^2||v||^2 + 2t < u, v > +||u||^2 \geqslant 0$

$\Delta = 4| < u, v > |^2 - 4||u||^2||v||^2 \leqslant 0$

$\therefore | < u, v > | \leqslant ||u||||v||$